

MATH 27300: Ordinary Differential Equations - Fall 2025 - midterm exam

Read and follow directions carefully.

- You have **80 minutes** to complete this midterm exam. Books, notes and electronic devices are not permitted.
- This exam contains **10** problems.
- The number of points for each problem is listed. Only this exam paper (including the **extra page**) will be collected and graded.
- For each problem, indexing your steps is strongly recommended to make sure you at least earn some partial score.
- This exam include 100 points and 14 pages.
- If you get stuck, state how you would proceed and move on to the next problem. If the instruction of the problem is not clear, you can follow the approach in the homework.
- You can use the back of the paper for scratch. Your draft **will not be graded**.
- **Hint:** **None** of the problems involve **lengthy** computation. If your derivation is quite long, you may get into the wrong direction.

Problem: 1	2	3	4	5	6	7	8	9	10	Total
Points: 15	15	10	7	10	13	10	10	5	5	100

I pledge that I have completed this midterm exam in accordance with the University of Chicago's policy on academic integrity. In particular, I have neither given nor received any unauthorized assistance during this exam.

Print Name: _____

Signature: _____

CNetID/UCMEDID : _____

Date: _____

Some formulas that might be useful

(a) Euler formula

$$e^{ix} = \cos x + i \sin x.$$

(b) Jordan normal form. Given a matrix $A \in \mathbb{C}^{n \times n}$. We can decompose A as

$$A = PJP^{-1}, \quad J = \begin{pmatrix} J_{k_1}(\lambda_1) & & & \\ & J_{k_2}(\lambda_2) & & \\ & & \ddots & \\ & & & J_{k_m}(\lambda_m) \end{pmatrix},$$

where $P \in \mathbb{C}^{n \times n}$ is an invertible matrix, λ_i is the eigenvalue of A , J_{k_i} is given by

$$J_{k_i}(\lambda_i) = \begin{pmatrix} \lambda_i & 1 & & & \\ & \lambda_i & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda_i & 1 \\ & & & & \lambda_i \end{pmatrix}_{k_i \times k_i}.$$

and $k_1 + k_2 + \dots + k_m = n$.

Problem 1. Basic concepts and short questions [$5 \times 3 = 15$ points, 3 % each]

- For the following questions, you only need to provide a short answer (≤ 20 words)
- You **do not** need to explain your solution.
- Answer **T** = true, **F** = False, or **STSF** = sometimes true and sometimes false.

(a) We can rewrite a $n - th$ order ODE for a scalar function y

$$\frac{d^n}{dt^n}y = F(t, y, \frac{d}{dt}y, \dots, \frac{d^{n-1}}{dt^{n-1}}y).$$

as first order ODE system.

T

(b) The following ODE has a unique solution near $(t_0 = 1, y_0 = 1)$.

$$\frac{d}{dt}y = |y - 1|^{1/2}(t - 1).$$

F

(c) If $y(t)$ solves a linear $n - th$ order homogeneous ODE, then $-y(t)$ also solves it. We **do not** consider initial condition.

T

(d) If $y_1(t)$ and $y_2(t)$ are two solutions of a second-order linear ODE, then any linear combination $c_1y_1(t) + c_2y_2(t)$ is also a solution.

STSF.

True : homogeneous ODE

False : Non homogeneous

(e) If $A, B \in \mathbb{R}^{n \times n}$ are $n \times n$ matrices, then $e^{A+B} = e^A e^B$, where e^A, e^B are the matrix exponentials.

STSF

True: when $AB = BA$

False: General $AB \neq BA$

Problem 2. Various methods [$5 \times 3 = 15$ points] (3 % each)

- State **Yes/No**: Can we obtain a general solution with course methods?
- If **Yes**, name the **ODE type** and the **solution method** (≤ 20 words).
- Do not solve the ODE.
- Implicit solutions count as “**obtainable.**”

Below, $y' = \frac{dy(t)}{dt}$.

- (a) $y' = e^t y + t^{-2025}$.

Y. First order Linear ODE.
Use integrating factor

- (b) $(t^2 + 1)y'' + ty' + y = \sin(t)$

Y. Power series + Variation of constants.

- (c)

$$\frac{d}{dt} \vec{X}(t) = A(t) \vec{X}(t),$$

where $A(t) \in \mathbb{R}^{n \times n}$ depends on t and $\vec{X} \in \mathbb{R}^{n \times 1}$.

No

- (d) $y'' + (y')^2 e^t = 0$.

Introduce $z = y'$

Y.

Get a 1st order ODE in z
Then use separation of variable

- (e) Suppose that the functions $M_i, N_i \neq 0$ and are smooth. Consider the ODE

$$M_1(x)N_1(y) + M_2(x)N_2(y)\frac{dy}{dx} = 0.$$

Y. Separation of variables.

Problem 3 [10 points] Consider the following non-homogeneous first order linear ODE

$$\frac{dy}{dt} = \sin(t)y + t.$$

(a) [5 points] Derive the integrating factor $\mu(t)$.

(b) [5 points] Further derive the general solution. You **do not** need to simplify the integral.

4. **Exact equations [7 points]** Find all functions $f(y, t)$ such that

$$(t-1)f(y, t) + ((t^3 - 3t)y + y^2) \frac{dy}{dt} = 0$$

is exact. You do not need to further solve the ODE.

Exact eqn $\Rightarrow$

$$\partial_y [(t-1)f(y, t)] = \partial_t [(t^3 - 3t)y + y^2]$$

Then solve on ODE on f .

Problem 5. 2nd order ODE [10 pts] (a) [5 points] Derive the general solutions $y = C_1 y_1(t) + C_2 y_2(t)$ to the following 2nd order constant linear ODE

$$y'' + 5y' + 6y = 0. \tag{1}$$

(b)[5 points] Suppose that $y_1(t), y_2(t)$ are solutions to (1). Consider

$$y'' + 5y' + 6y = e^t.$$

with solution $y(t) = \mu_1(t)y_1(t) + \mu_2(t)y_2(t)$. Derive the formula for $\mu_1'(t), \mu_2'(t)$. You do not need to further derive the formula of μ_1, μ_2 and $y(t)$.

Problem 6. Series solution [13 points] Consider the following ODE

$$y''(t) + ty' + k^2 y(t) = 0.$$

where $k > 0$ is a parameter. We consider solving the above ODE near $t = 0$.

(a) [6 points] Derive the power series expansion

$$y(t) = \sum_{n \geq 0} a_n t^n$$

and the equations of the coefficients a_n .

(b) [7 points] Find out two linearly independent solutions by obtaining the formulas of a_n . If you cannot obtain the closed form of a_n , you need to provide the first 4 terms in each of the series solution.

Extra page for Problem 6.

Problem 7. Linear system [10 points]

(1) [5 points] Derive the general solutions and the fundamental matrix solution e^{At} of the following ODE

$$\frac{d}{dt}X = AX, \quad A = \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix}.$$

You can use the method of Jordan normal form.

(2) [5 points] Solve the following non-homogeneous equation

$$\frac{d}{dt}X = \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix} X + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \sin(t), \quad X_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

using e^{At} derived in (1). You *do not* need to simplify the integral.

Extra space for problem 7.

Problem 8. Growth of the solution [10 pts] Consider the ODE

$$\frac{d}{dt}y = f(t, y), \quad y(0) = 1.$$

Suppose that there exists constant $A, B > 0$ such that the function f satisfies

$$|f(t, y)| \leq A|y| + B.$$

for any $(t, y) \in \mathbb{R} \times \mathbb{R}$, it is continuous and Lipschitz everywhere. Suppose that the solution exists on $t \in [0, T]$. Use Gronwall's inequality or comparison principle to prove that

$$y(t) \leq e^{At} + \frac{B}{A}(e^{At} - 1), \quad \forall t \in [0, T].$$

See HW

Problem 10. [5 pts] Consider the following **second** order linear system.

$$\frac{d^2}{dt^2}X(t) = A(t)X(t), \quad X(0) = X_0, \quad \frac{d}{dt}X(0) = X_1.$$

where $X(t), X_0, X_1 \in \mathbb{R}^{n \times 1}$ and $A(t) \in \mathbb{R}^{n \times n}$. Since the ODE is second order, we provide initial data for $X(0)$ and $\frac{d}{dt}X(0)$. Suppose that the matrix A satisfies

$$\max(|A_{ij}(t)|) \leq a, \quad \forall t \geq 0.$$

for some parameter $a > 0$ independent of t . Suppose that the solution $X(t)$ exists on $t \in [0, T]$. Prove that there exists an absolute constant C_n , that only depends on n , but independent of a, A, X_0, X_1, t, T , such that

$$\max_i |X_i(t)| \leq C_n e^{C_n(a+1)t} \max_i (|X_{0,i}| + |X_{1,i}|).$$

• Rewrite it as

$$\frac{d}{dt} \begin{pmatrix} X(t) \\ X'(t) \end{pmatrix} = \begin{pmatrix} 0 & I \\ A & 0 \end{pmatrix} \begin{pmatrix} X(t) \\ X'(t) \end{pmatrix}$$

$$\gamma(t) \stackrel{\circ}{=} \begin{pmatrix} X(t) \\ X'(t) \end{pmatrix}, \quad \gamma(0) = \begin{pmatrix} X(0) \\ X'(0) \end{pmatrix}$$

• Apply Gronwall's inequality to $\gamma(t)$. $\square$